THE VIRTUAL MEET EXPERIENCE

2025-2026

HS VIRTUAL CHALLENGE MEET #4

COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

The Virtual Challenge Meets™

2025-2026 High School Virtual Challenge Meet #4

General Directions:

1) DO NOT OPEN EXAM UNTIL TOLD TO DO SO.
2) **NO CALCULATORS of any kind may be used.**
3) You have 45 minutes to complete this contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
4) Papers may not be turned in until forty-five minutes have elapsed. If you finish the test before the end of the allotted time, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
5) All answers must be written on the answer sheet/Scantron card provided. Indicate your answers in the appropriate blanks provided on the answer sheet or on the Scantron card. Clean erasures are necessary for accurate Scantron grading.
6) You may place as many notations as you desire anywhere on the test paper except on the answer sheet or Scantron card which is reserved for answers only.
7) You may use additional scratch paper provided by the contest director.
8) All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers. **All provided code segments are intended to be syntactically correct, unless otherwise stated (i.e. `error` is an answer choice). Ignore any typographical errors and assume any undefined variables are defined as used.**
9) A reference to commonly used Java classes is provided with the test and you may use this reference during the contest. You may detach the reference sheets from the test booklet but DO NOT DO SO UNTIL THE CONTEST BEGINS.
10) Assume that any necessary import statements for **Java 23** and classes (e.g. .lang, .util, System, Math, Double, etc.) are included in any programs or code segments that refer to methods from these classes and/or packages.

Scoring:

1) All questions will receive 6 points if answered correctly; no points will be given or subtracted if unanswered; 2 points will be deducted for each incorrect answer.
2)

For Computer Science practice tests and hands-on materials, go to www.apluscompsci.com

Standard Classes and Interfaces — Supplemental Reference

class java.lang.Object

- `boolean equals(Object other)`
- `String toString()`
- `int hashCode()`

interface java.lang.Comparable<T>

- `int compareTo(T other)`
  Return value < 0 if `this` is less than `other`.
  Return value = 0 if `this` is equal to `other`.
  Return value > 0 if `this` is greater than `other`.

class java.lang.Integer implements Comparable<Integer>

- `Integer(int value)`
- `int intValue()`
- `boolean equals(Object obj)`
- `String toString()`
- `int compareTo(Integer anotherInteger)`
- `static int parseInt(String s)`

class java.lang.Double implements Comparable<Double>

- `Double(double value)`
- `double doubleValue()`
- `boolean equals(Object obj)`
- `String toString()`
- `int compareTo(Double anotherDouble)`
- `static double parseDouble(String s)`

class java.lang.String implements Comparable<String>

- `int compareTo(String anotherString)`
- `boolean equals(Object obj)`
- `int length()`
- `String substring(int begin, int end)`
  Returns the substring starting at index `begin` and ending at index `(end - 1)`.
- `String substring(int begin)`
  Returns `substring(from, length())`.
- `int indexOf(String str)`
  Returns the index within this string of the first occurrence of `str`. Returns `-1` if `str` is not found.
- `int indexOf(String str, int fromIndex)`
  Returns the index within this string of the first occurrence of `str`, starting the search at the specified index.. Returns `-1` if `str` is not found.
- `charAt(int index)`
- `int indexOf(int ch)`
- `int indexOf(int ch, int fromIndex)`
- `String toLowerCase()`
- `String toUpperCase()`
- `String[] split(String regex)`
- `boolean matches(String regex)`

class java.lang.Character

- `static boolean isDigit(char ch)`
- `static boolean isLetter(char ch)`
- `static boolean isLetterOrDigit(char ch)`
- `static boolean isLowerCase(char ch)`
- `static boolean isUpperCase(char ch)`
- `static char toUpperCase(char ch)`
- `static char toLowerCase(char ch)`

class java.lang.Math

- `static int abs(int a)`
- `static double abs(double a)`
- `static double pow(double base, double exponent)`
- `static double sqrt(double a)`
- `static double ceil(double a)`
- `static double floor(double a)`
- `static double min(double a, double b)`
- `static double max(double a, double b)`
- `static int min(int a, in b)`
- `static int max(int a, int b)`
- `static long round(double a)`
- `static double random()`
  Returns a `double` value with a positive sign, greater than or equal to `0.0` and less than `1.0`.

interface java.util.List<E>

- `boolean add(E e)`
- `int size()`
- `Iterator<E> iterator()`
- `ListIterator<E> listIterator()`
- `E get(int index)`
- `E set(int index, E e)`
  Replaces the element at `index` with the object `e`.
- `void add(int index, E e)`
  Inserts the object `e` at position `index`, sliding elements at position `index` and higher to the right (adds 1 to their indices) and adjusts `size`.
- `E remove(int index)`
  Removes element from position `index`, sliding elements at position `(index + 1)` and higher to the left (subtracts 1 from their indices) and adjusts `size`.

class java.util.ArrayList<E> implements List<E>

class java.util.LinkedList<E> implements List<E>, Queue<E>

Methods in addition to the `List` methods:

- `void addFirst(E e)`
- `void addLast(E e)`
- `E getFirst()`
- `E getLast()`
- `E removeFirst()`
- `E removeLast()`

class java.util.Stack<E>
- `boolean isEmpty()`
- `E peek()`
- `E pop()`
- `E push(E item)`

interface java.util.Queue<E>
- `boolean add(E e)`
- `boolean isEmpty()`
- `E peek()`
- `E remove()`

class java.util.PriorityQueue<E>
- `boolean add(E e)`
- `boolean isEmpty()`
- `E peek()`
- `E remove()`

interface java.util.Set<E>
- `boolean add(E e)`
- `boolean contains(Object obj)`
- `boolean remove(Object obj)`
- `int size()`
- `Iterator<E> iterator()`
- `boolean addAll(Collection<? extends E> c)`
- `boolean removeAll(Collection<?> c)`
- `boolean retainAll(Collection<?> c)`

class java.util.HashSet<E> implements Set<E>

class java.util.TreeSet<E> implements Set<E>

interface java.util.Map<K,V>
- `Object put(K key, V value)`
- `V get(Object key)`
- `boolean containsKey(Object key)`
- `int size()`
- `Set<K> keySet()`
- `Set<Map.Entry<K, V>> entrySet()`

class java.util.HashMap<K,V> implements Map<K,V>

class java.util.TreeMap<K,V> implements Map<K,V>

interface java.util.Map.Entry<K,V>
- `K getKey()`
- `V getValue()`
- `V setValue(V value)`

interface java.util.Iterator<E>
- `boolean hasNext()`
- `E next()`
- `void remove()`

interface java.util.ListIterator<E> extends java.util.Iterator<E>

Methods in addition to the `Iterator` methods:
- `void add(E e)`
- `void set(E e)`

class java.lang.Exception
- `Exception()`
- `Exception(String message)`

class java.util.Scanner
- `Scanner(InputStream source)`
- `boolean hasNext()`
- `boolean hasNextInt()`
- `boolean hasNextDouble()`
- `String next()`
- `int nextInt()`
- `double nextDouble()`
- `String nextLine()`
- `Scanner useDelimiter(String pattern)`

2025-2026 Virtual Challenge Meet #4 – Computer Science

Note: Correct responses are based on **Java 23**. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported using:** ***import static java.lang.System.*;***

QUESTION 1

Which of the choices below is the equivalent to 1111_2 ?

A. 10000_2 B. 14_{10} C. 15_8 D. 22_3 E. F_{16}

QUESTION 2

What is output by the code to the right?

A. 7 B. 28
C. 30 D. 54
E. 32

```
out.println(24 + 32 / 8);
```

QUESTION 3

What is output by the code to the right?

A. 9.44AP B. 9.45AP
C. 9.43AP D. 9.44
E. There is no output due to a syntax error.

```
out.printf("%.2f%.2s",9.4358,"APLUS");
```

QUESTION 4

What is output by the code to the right?

A. aplus5 B. aplusCS
C. aplusCS7 D. CS2
E. aplusCSCS7

```
String a = "aplus";
String b = "CS";
a += b;
out.println(a + b + a.length());
```

QUESTION 5

What is output by the code to the right?

A. true B. false

```
out.println(false | true & false );
```

QUESTION 6

What is output by the code to the right?

A. 81 B. 81.0
C. 27 D. 27.0
E. There is no output due to a syntax error.

```
out.println(Math.pow(3,4));
```

QUESTION 7

What is output by the code to the right?

A. 5.5 B. 2.5
C. 2.0 D. 2
E. 3.0

```
int i = 4, n=2;
double d = 1.5;
out.println( i * ( n - d ));
```

QUESTION 8

What is output by the code to the right?

A. 3 B. 4 C. 35

D. 34 E. 45

```
int x = 10, z = 7;
if(x > z)
  out.print("3");
else
  out.print("4");
out.print("5");
```

QUESTION 9

What is output by the code to the right?

A. 7131925 B. 11713192

C. 17131925 D. 17131926

E. 171319

```
for(int i = 1; i <= 25; i += 4){
   out.print(i);
   i+=2;
}
```

QUESTION 10

What is output by the code to the right?

A. PLA B. APL C. 212 D. 221

E. There is no output due to a syntax error.

```
char ray[] = "APLUSCOMPSCI".toCharArray();
out.print(ray[1] + ray[2] + ray[0]);
```

QUESTION 11

What is output by the code to the right?

A. APLUS B. COMP

C. SCI D. COMPSCI

E. APLUSCOMPSCI

```
String s = "www.apluscompsci.com";
s = "APLUS COMP SCI";
Scanner in = new Scanner(s);
in.next();
out.println(in.next());
in.next();
```

QUESTION 12

What is output by the code to the right?

A. 5 B. -6

C. -3 D. 0

E. There is no output due to a runtime error.

```
int x = 50;
while(x > 0){
  x = x - 7;
}
out.print(x);
```

QUESTION 13

What is output by the code to the right?

A. 9 B. 24

C. 27 D. 14

E. 13

```
out.println(9 | 5 ^ 13);
```

QUESTION 14

Given the assignments to the right, which of the following statements below will NOT produce an *incompatible types* error ?

A. s = i; B. l = d;

C. d = l; D. s = c;

E. i = d;

```
short s = 217;
int i = 212212;
char c = 'z';
double d = 212.212;
long l = 212;
```

QUESTION 15

What is output by the code to the right?

A. 7 B. -4
C. 3 D. 2
E. null

```
ArrayList<Integer> r;
r = new ArrayList<>();
r.add(2);
r.add(3);
r.add(-4);
r.add(7);
Collections.reverse(r);
out.print(r.remove(0));
```

QUESTION 16

What is output by `//line 1` in the code to the right?

A. G B. GU
C. UG D. U
E. GG

QUESTION 17

What is output by `//line 2` in the code to the right?

A. GU B. GK
C. UK D. K
E. GUK

QUESTION 18

What is output by `//line 3` in the code to the right?

A. GU B. GK
C. UK D. K
E. GUK

```
class Z{
   {  out.print("G");   }
   Z(){ out.print("U"); }
}

class Y extends Z{
   Y(){  out.print("K");  }
}

//////////////////////////////////////
//client code
Z r = new Z();    //line 1
out.println();
Y b = new Y();    //line 2
out.println();
Z w = new Y();    //line 3
```

QUESTION 19

What is output by the code to the right?

A. 010 B. 10
C. 48 D. 58
E. There is no output due to a syntax error.

```
out.println('0'+10);
```

QUESTION 20

What is output by the code to the right?

A. 8 B. 25
C. 29 D. 14
E. 10

```
String p = "APLUSAPLUSAPLUSAPLUSAPLUS";
p = p.replaceAll("[APS]","");
out.println(p.length());
```

QUESTION 21

Which keyword is required in place of **<code1>** to make the method `aplus` work properly?

A. `int[]`
B. `int`
C. `private`
D. `double[]`
E. `void`

QUESTION 22

Assuming **<code1>** has been filled correctly, what is output by the code to the right?

A. `5 5 5`
B. `0 5 1`
C. `0 5 5`
D. `5 5 0`
E. `5 5 1`

```
public static <code1> aplus(int[] r)
{
   int[] t = r;
   if(r[0] == 0)
      t[0] = 5;
   else
      r[0] = 5;
   out.print(r[0] + " ");
   out.print(t[0] + " ");
}


////////////////////////////////////
//client code
int[] t = {1,2,3,4};
aplus(t);
System.out.println(t[0]);
```

QUESTION 23

What is output by the code to the right?

A. `12`
B. `11`
C. `13`
D. `10`
E. `14`

```
int a = 10;
a = ++a;
a = a++;
out.println(++a);
```

QUESTION 24

What is output by the code to the right?

A. `true`
B. `false`

```
String a = "X";
String z = new String("X");
String w = "X";
out.print(a == z ^ a == w);
```

QUESTION 25

What is output by the code to the right?

A. `4`
B. `8`
C. `16`
D. `2`
E. `20`

```
out.println(128 >> 15 % 4 + 1);
```

QUESTION 26

What is output by the code to the right?

A. `44`
B. `54`
C. `72`
D. `62`
E. `50`

```
int z = 0;
for( int a = 1; a < 5; a++)
  for( int b = 1; b <= a; b++ )
  {
     z += a + b;
  }
out.println( z );
```

QUESTION 27

What is output by `//line 1` in code to the right?

A. `[-10, -5, 2, 27, 17, 10, 7]`

B. `[-10, 2, -5, 27, 7, 10, 17]`

C. `[-10, -5, 2, 7, 10, 17, 27]`

D. `[-10, 2, -5, 27, 17, 10, 7]`

E. `[-10, 2, -5, 17, 7, 10, 27]`

QUESTION 28

What is output by `//line 2` in code to the right?

A. `[2, 10, 7, 27, 17]`

B. `[2, 7, 10, 27, 17]`

C. `[2, 7, 10, 17, 27]`

D. `[7, 10, 2, 27, 17]`

E. `[27, 7, 10, 2, 17]`

```
PriorityQueue<Integer> pq;
pq = new PriorityQueue<>();
pq.add(17);
pq.add(7);
pq.add(-5);
pq.add(27);
pq.add(2);
pq.add(10);
pq.add(-10);
out.println(pq);   //line 1
pq.remove();
pq.remove();
out.println(pq);   //line 2
```

QUESTION 29

What is returned by the call `po(3, 22)`?

A. `24`

B. `6`

C. `12`

D. `36`

E. `48`

```
public static int po(int x, int y)
{
  if(x < y)
    return po(x+2,y-5)+po(x+3,y-4);
  else
    return 6;
}
```

QUESTION 30

What is output by the code to the right?

A. `CAT dog up 22`

B. `dog up CAT 22`

C. `22 dog up CAT`

D. `up CAT dog 22`

E. `22 CAT dog up`

```
ArrayList<String> list;
list = new ArrayList<>();
list.add("dog");
list.add("CAT");
list.add(1,"pig");
list.add(1,"up");
list.remove(2);
list.add(1,"22");
Collections.sort(list);
for(String s: list)
  out.print(s + " ");
```

QUESTION 31

The following values are entered into a binary tree in the order listed: `14 24 15 8 27 2 10`

Which of the following represents the values of a post-order traversal of this tree?

A. `2 10 8 15 27 24 14`

B. `2 8 10 14 15 24 27`

C. `14 8 2 10 24 15 27`

D. `27 24 15 14 10 8 2`

E. `27 15 10 2 24 8 14`

QUESTION 32

What is output by the code to the right?

A. 34

B. 4

C. 17

D. 25

E. 30

```
int x = Integer.parseInt("34",7);
out.println(x);
```

QUESTION 33

If a balanced binary search tree has 25000 elements, what is the maximum number of checks needed to find any value in the tree?

A. 25000 B. 12500 C. 15 D. 12 E. 18

QUESTION 34

What is output by the code to the right?

A. 33

B. 16

C. 3

D. 7

E. There is no output due to a syntax error.

```
String s = "apluscompsci.com";
s += "practice.apluscompsci.com";
String ray[] = s.split("[^apluscompsci]");
out.println(ray.length);
```

QUESTION 35

What is output by `//line 1` code to the right?

A. 4 B. 3

C. -8 D. 33

E. 11

QUESTION 36

What is output by `//line 2` code to the right?

A. [3, 11, 5, 33]

B. [33, 5, 11, 3]

C. [-8, 4, 33, 11]

D. [33, 11, 5, 3]

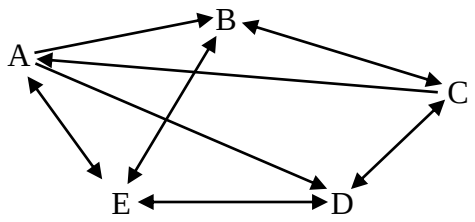
E. [3, 5, 11, 33]

```
Stack<Integer> s;
s = new Stack<>();

s.add(3);
s.push(5);
s.add(11);
s.push(33);
s.add(4);
s.push(-8);
out.println(s.pop());      //line 1


s.pop();
s.peek();
out.println(s);       //line 2
```

QUESTION 37

How many cycles exist in the graph to the right that start and end at vertex A?

A. 3
B. 4
C. 5
D. 7
E. 0



QUESTION 38

Which of the following is the result of converting the infix expression to the right to postfix?

A. G O O ^ G * / L E * +
B. G O ^ O G * / L E * +
C. G O ^ O G / L E * * +
D. G ^ O O * G / L E * +
E. + * G O / * O G ^ L E

G ^ O / (O * G) + L * E

QUESTION 39

Which data structure is most commonly used to implement a priority queue?

QUESTION 40

Name one of the algorithms commonly used to implement the behavior of the ghosts in Pac-Man.

Name _____

School _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12

Final Score _____ _____

2nd Grading _____ _____

1st Grading _____ _____
score initials

The Virtual Challenge Meets
Computer Science - Student Answer Sheet

1) _____
2) _____
3) _____
4) _____
5) _____
6) _____
7) _____
8) _____
9) _____
10) _____

11) _____
12) _____
13) _____
14) _____
15) _____
16) _____
17) _____
18) _____
19) _____
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21) _____
22) _____
23) _____
24) _____
25) _____
26) _____
27) _____
28) _____
29) _____
30) _____

31) _____
32) _____
33) _____
34) _____
35) _____
36) _____
37) _____
38) _____
39) _____
40) _____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

2025-2026
Virtual Challenge Meet #4
Computer Science - KEY

1) E
2) B
3) A
4) E
5) B
6) B
7) C
8) C
9) C
10) D

11) B
12) B
13) A
14) C
15) A
16) B
17) E
18) E
19) D
20) E

21) E
22) A
23) A
24) A
25) B
26) E
27) D
28) A
29) E
30) E

31) A
32) D
33) C
34) D
35) C
36) E
37) C
38) B
39) heap, max-heap, min-heap, or binary-heap
40) BFS, DFS, A*search, or Dijkstra

Note to Graders:

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors**.
- Any necessary Standard Java 23 Packages are assumed to have been imported as needed.
- Assume any undefined (undeclared) variables have been defined as used.

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